

MET CS 555 Assignment 1 – 20 points

SUBMISSION REQUIREMENTS: Please submit a single document (word or PDF) for submission. Your submission should contain a summary of your results (and answers to questions asked on the homework) as well as your R code used to generate your results (**please append your R code to the end of your submission**).

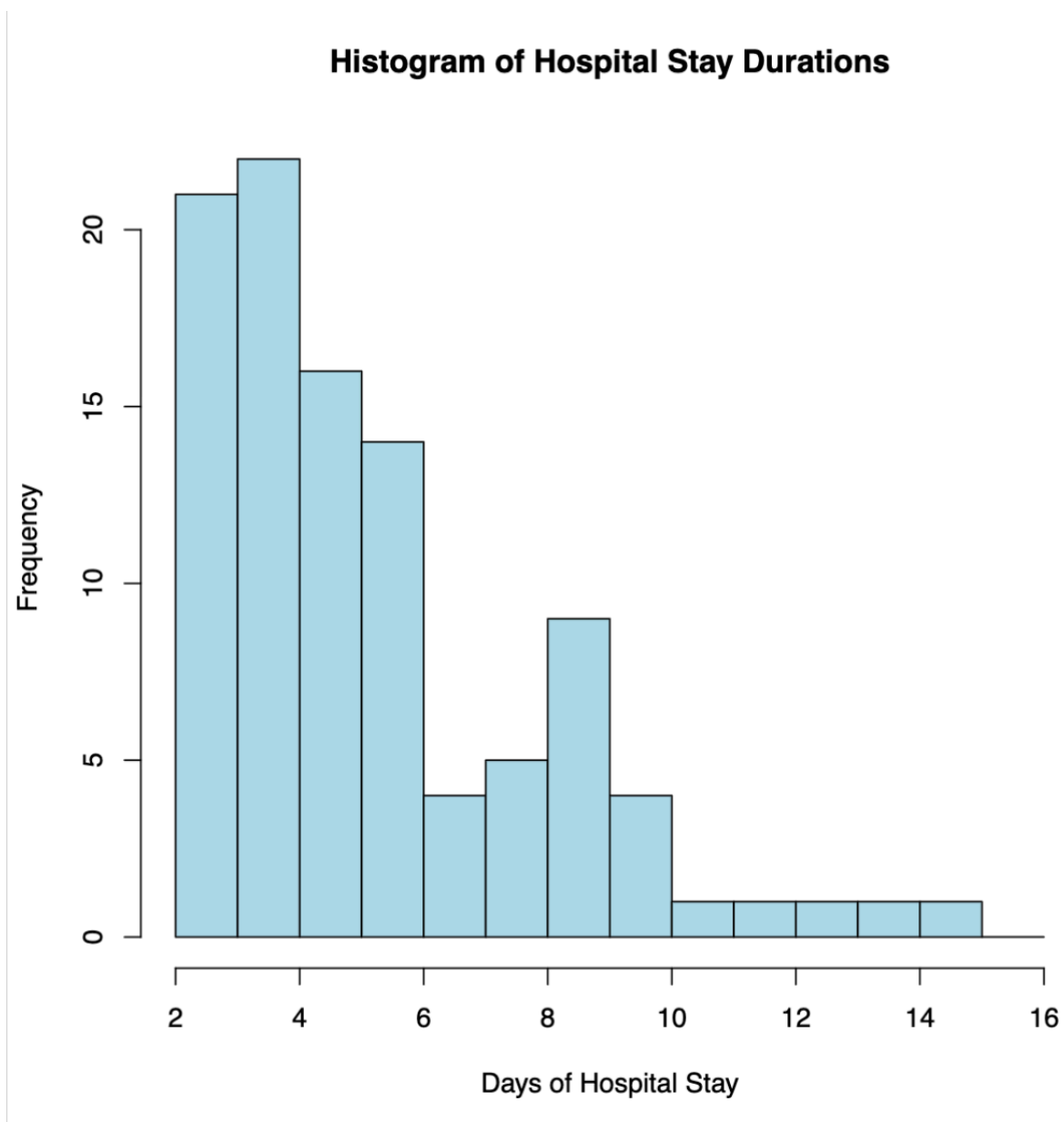
Homework submission filenames should take on the form **LASTNAME - Homework #.doc(x)/.pdf**.

No ZIP files please.

The data in the table below give the duration in days of hospital stays of patients admitted to the hospital with *C. Difficile*. Use the data on the following page to:

Note that the data is one column, but the long column has been broken by 10 rows each and they attached to each other in a form table below.

- (1) Save the data to a CSV file and read into R for analysis. **(2 points)**
- (2) Make a histogram of the duration of days of hospital stays. Ensure the histogram is labelled appropriately. Use a width of **1 day**. Describe the shape, center, and spread of the data. Are there any outliers? **(5 points)**



The histogram is right-skewed, the center is around 5, and the spread is from 2 - 15.

Outliers: 15 13 14 12

(3) Find the mean, median, standard deviation, first and third quartiles, minimum and maximum of the durations of hospital stay in the sample. **Summarize these values in a table that you create in EXCEL or WORD.** In other words, do **not** simply copy and paste R output. You should be reporting a nicely labeled and formatted table.

Statistic	Value
Mean	5.63
Median	5
Standard Deviation	2.74
Q1	4
Q3	7
Minimum	2
Maximum	15

- Given the shape of the distribution, what is the best single number summary of the center of the distribution?

5, because the outliers inflate the mean so we use the median.

- What is the best single number summary of the spread of the distribution? **(5 points)**

3, we use the interquartile value because of the outliers.

(4) Assume that the literature on this topic suggests that the distribution of days of hospital stay are normally distributed with **a mean of 5** and **a standard deviation of 3**. Use R to determine the probabilities below based on the normal distribution described above (you should **not** be using the data set given on the following page):

(a) What percentage of patients are in the hospital for less than **10 days**? **(2 points)**

95.22 %

What percentage of patients are in the hospital between **3 and 10 days**? **(2 points)**

69.97 %

(b) Recent publications have indicated that hypervirulent strains of *C. Difficile* are on the rise. Such strains are associated with poor outcomes, including extended hospital stays. An investigator is interested in showing that the average hospital stay duration have increased versus published literature. He has a sample of **35 patients** from his hospital. If the published data are consistent with the truth, what is the probability that the sample mean in his sample will be greater than **6 days**? (4 points)

2.4%

Data Set for Assignment 1

7	3	5	3	9	5	10	3	4	4
7	5	8	3	4	9	15	4	5	8
5	3	2	3	5	9	4	5	6	9
5	3	6	3	2	6	4	5	5	4
5	8	4	6	13	4	6	3	2	3
2	4	6	6	6	8	6	3	4	4
5	10	4	6	3	9	3	9	4	7
10	14	4	6	5	10	4	4	9	4
4	3	6	8	5	7	6	9	3	12
11	5	2	9	4	4	5	6	4	2

```
# Read CSV file into a data frame

data = read.csv("data.csv", header = FALSE)

first_col = data[[1]]

head(first_col)
```

```
# Make a histogram of hospital stay durations

hist(first_col,

      breaks = seq(min(first_col),

                    max(first_col) + 1,

                    by = 1),

      main = "Histogram of Hospital Stay Durations",

      xlab = "Days of Hospital Stay",

      ylab = "Frequency",

      col = "lightblue",

      border = "black")

# Compute quartiles and IQR

Q1 <- quantile(first_col, 0.25, na.rm = TRUE)

Q3 <- quantile(first_col, 0.75, na.rm = TRUE)

IQR_value <- IQR(first_col, na.rm = TRUE)

# Define outlier bounds
```

```
lower_bound <- Q1 - 1.5 * IQR_value  
upper_bound <- Q3 + 1.5 * IQR_value  
  
# Identify outliers  
outliers <- first_col[first_col < lower_bound |  
first_col > upper_bound]  
  
# Print results  
cat("Q1:", Q1, "\n")  
cat("Q3:", Q3, "\n")  
cat("IQR:", IQR_value, "\n")  
cat("Lower bound:", lower_bound, "\n")  
cat("Upper bound:", upper_bound, "\n")  
cat("Outliers detected:", outliers, "\n")  
  
# Summary statistics  
mean_val <- mean(first_col, na.rm = TRUE)  
median_val <- median(first_col, na.rm = TRUE)
```

```
sd_val <- sd(first_col, na.rm = TRUE)

min_val <- min(first_col, na.rm = TRUE)

max_val <- max(first_col, na.rm = TRUE)


cat("\nSummary Statistics:\n")

cat("Mean:", mean_val, "\n")

cat("Median:", median_val, "\n")

cat("Standard Deviation:", sd_val, "\n")

cat("Q1:", Q1, "\n")

cat("Q3:", Q3, "\n")

cat("Minimum:", min_val, "\n")

cat("Maximum:", max_val, "\n")


# Create summary table

summary_table <- data.frame(

  Statistic = c("Mean", "Median", "Standard
Deviation", "Q1", "Q3", "Minimum", "Maximum"),

  Value = c(mean_val, median_val, sd_val, Q1, Q3,
min_val, max_val)
```

```
)

print(summary_table, row.names = FALSE)

# Probabilities based on normal distribution  $N(5, 3^2)$ 

mean_norm <- 5

sd_norm <- 3

# (a)  $P(X < 10)$ 

p_less_10 <- pnorm(10, mean = mean_norm, sd =
sd_norm)

# (b)  $P(3 \leq X \leq 10)$ 

p_between_3_10 <- pnorm(10, mean = mean_norm, sd =
sd_norm) - pnorm(3, mean = mean_norm, sd = sd_norm)
```



```
cat("\nProbabilities (Normal distribution
N(5,3^2)):\n")

cat("Percentage of patients with stays < 10 days:",
round(p_less_10 * 100, 2), "%\n")

cat("Percentage of patients with stays between 3
and 10 days:", round(p_between_3_10 * 100, 2),
"%\n")

# last question

mu = 5

sigma = 3

n = 35

sigma_clt = 3 / sqrt(35)

z_score = (6 - mu) / sigma_clt
```

```
p = pnorm(z_score, lower.tail = FALSE)

cat("The probability of observing a sample with
mean > 6 is ", p)
```